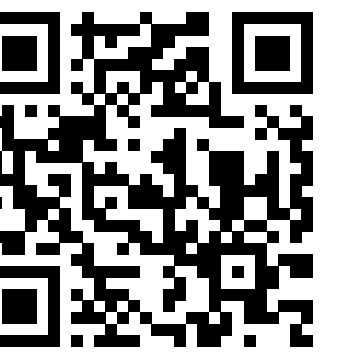
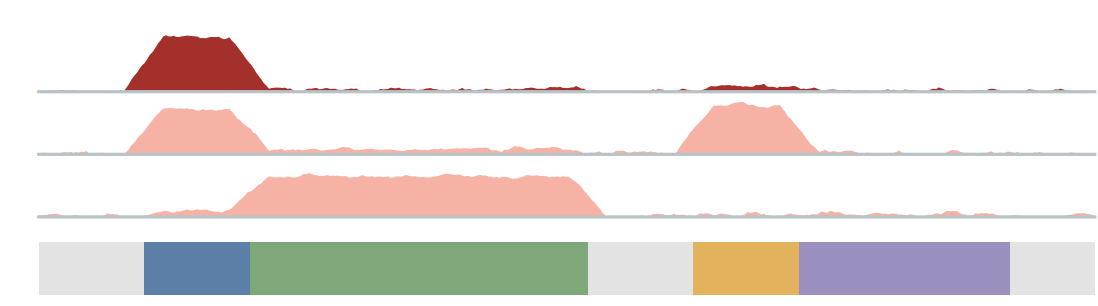




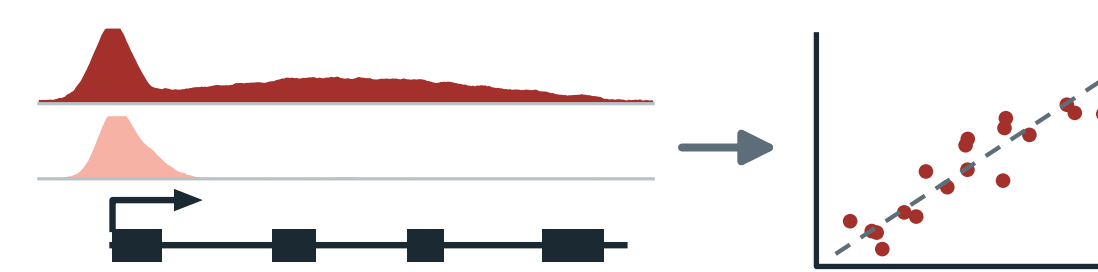
## Epigenome Imputation problem

- Epigenomic data (e.g., histone modifications, DNA accessibility) span tens to hundreds of assays and thousands of cell types.
- Due to the high cost and complexity of profiling every assay in all relevant cell types, most experiments are missing.
- Epigenome imputation methods aim to predict these missing experiments [1,2].
- Imputed data often has better properties than observed data, so researchers also apply imputation for denoising [1].
- In the ENCODE Imputation Challenge, due to subtle differences in experimental covariates between the train and test sets, a simple average-activity baseline outperformed most of the entrants [2].

## Downstream applications of imputation



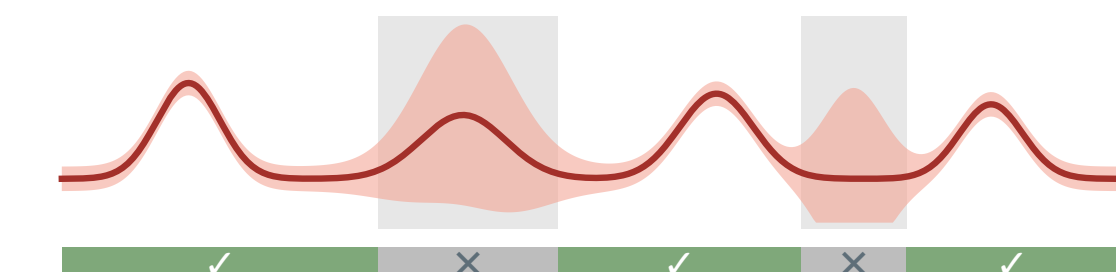
Annotate chromatin states in every cell type [1].



Predict gene expression from marks around each gene [1,3].

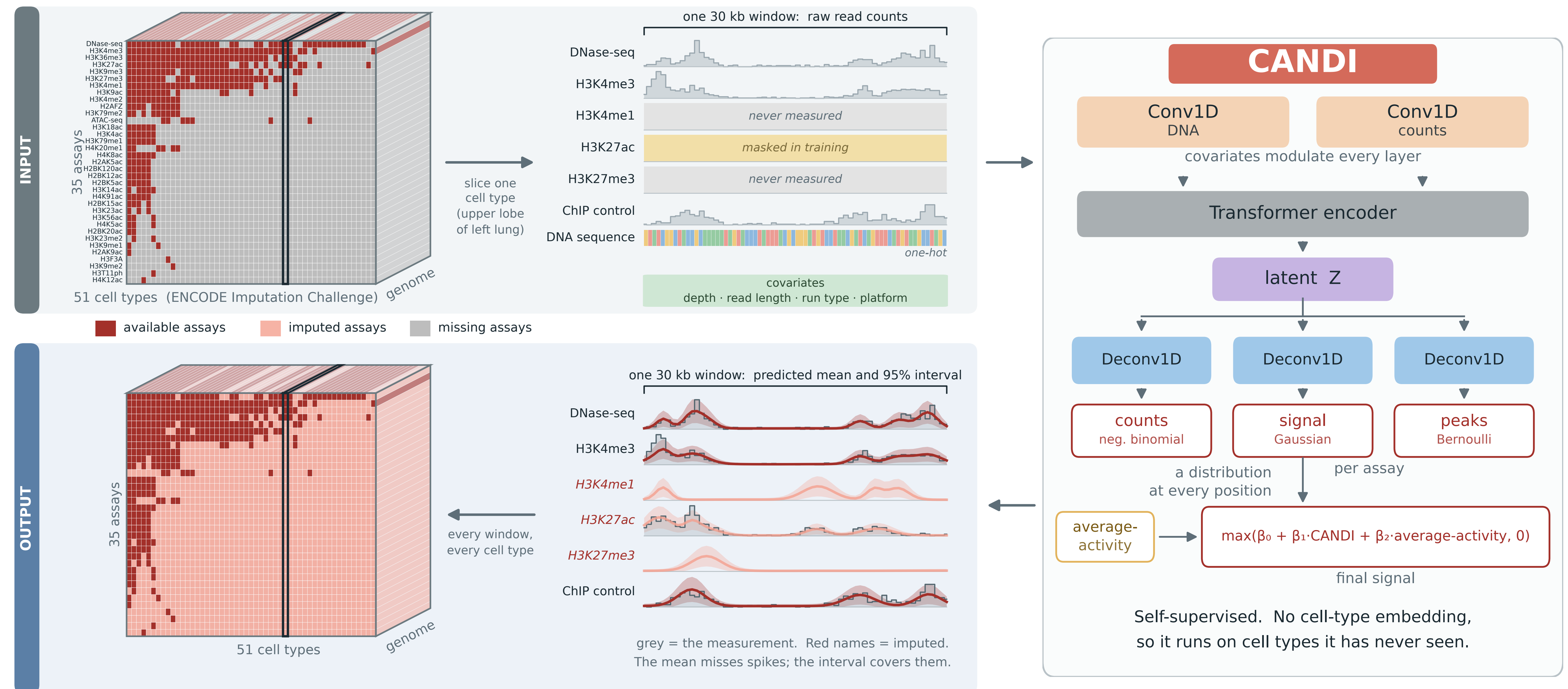


Link disease-associated variants to cell type-specific regulatory elements [1].



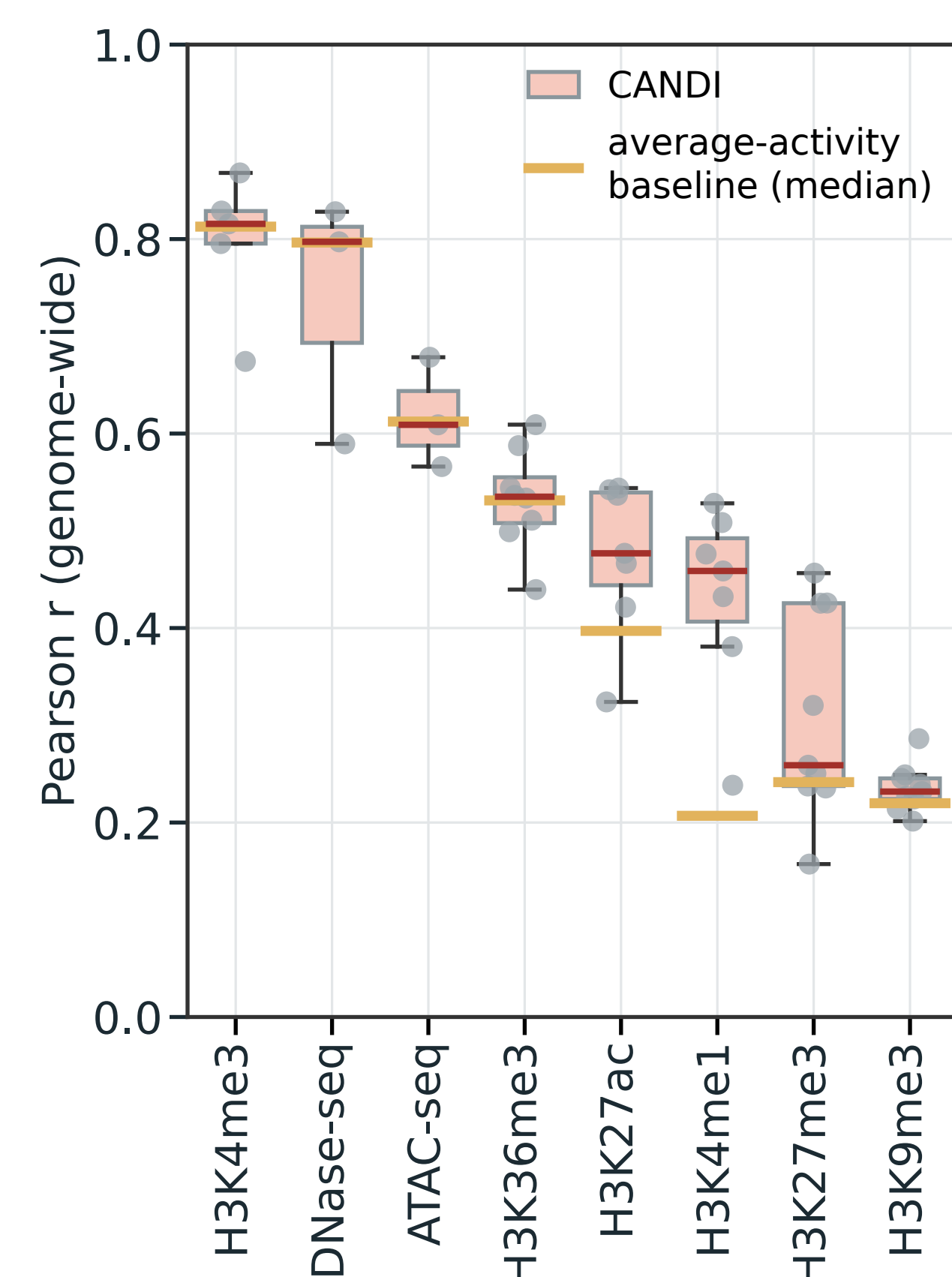
Keep confident predictions; discard uncertain predictions [3].

## CANDI schematic workflow

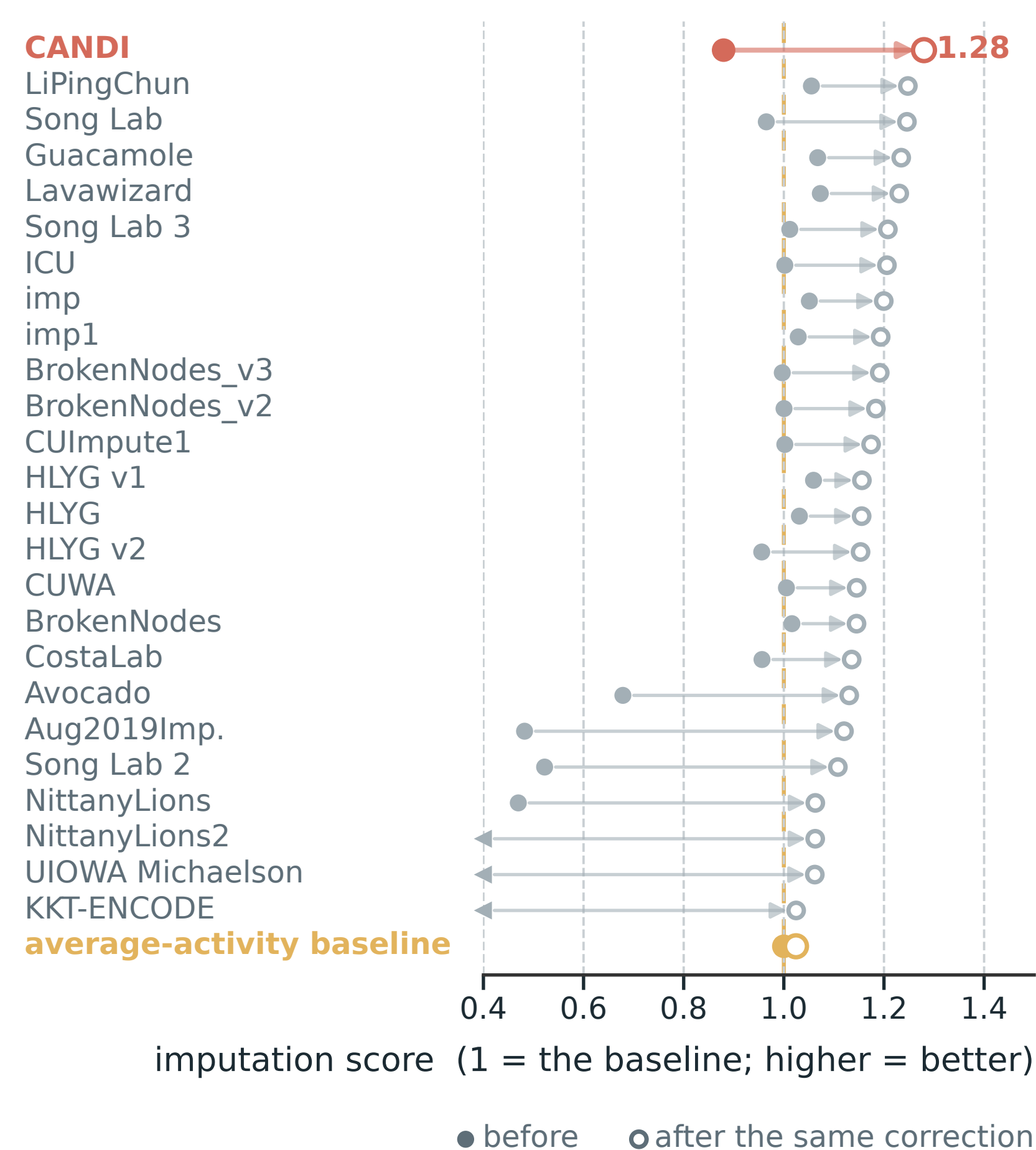


**Top:** for each 30 kb locus, CANDI takes observed read counts with the ChIP-seq control, four covariates per assay and the DNA sequence; grey assays were never measured, yellow ones are masked. **Bottom:** each output is a probability distribution, here a mean and 95% interval per assay; window by window, they fill every missing thread of the tensor.

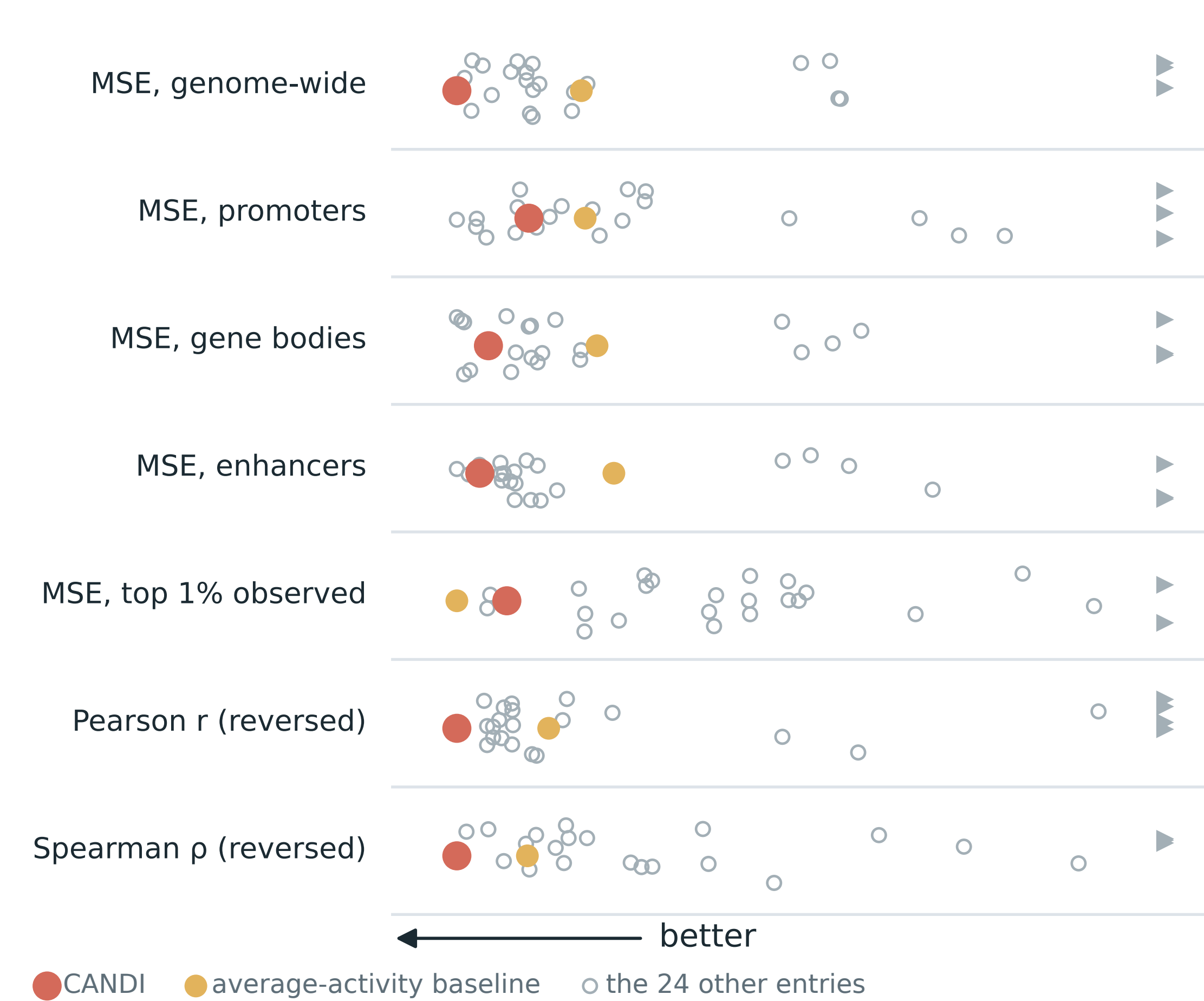
## Fixing the train-test covariate shift in the ENCODE Imputation Challenge



**A** CANDI's genome-wide Pearson  $r$  on the 51 blind-test experiments; mustard: the baseline's median.

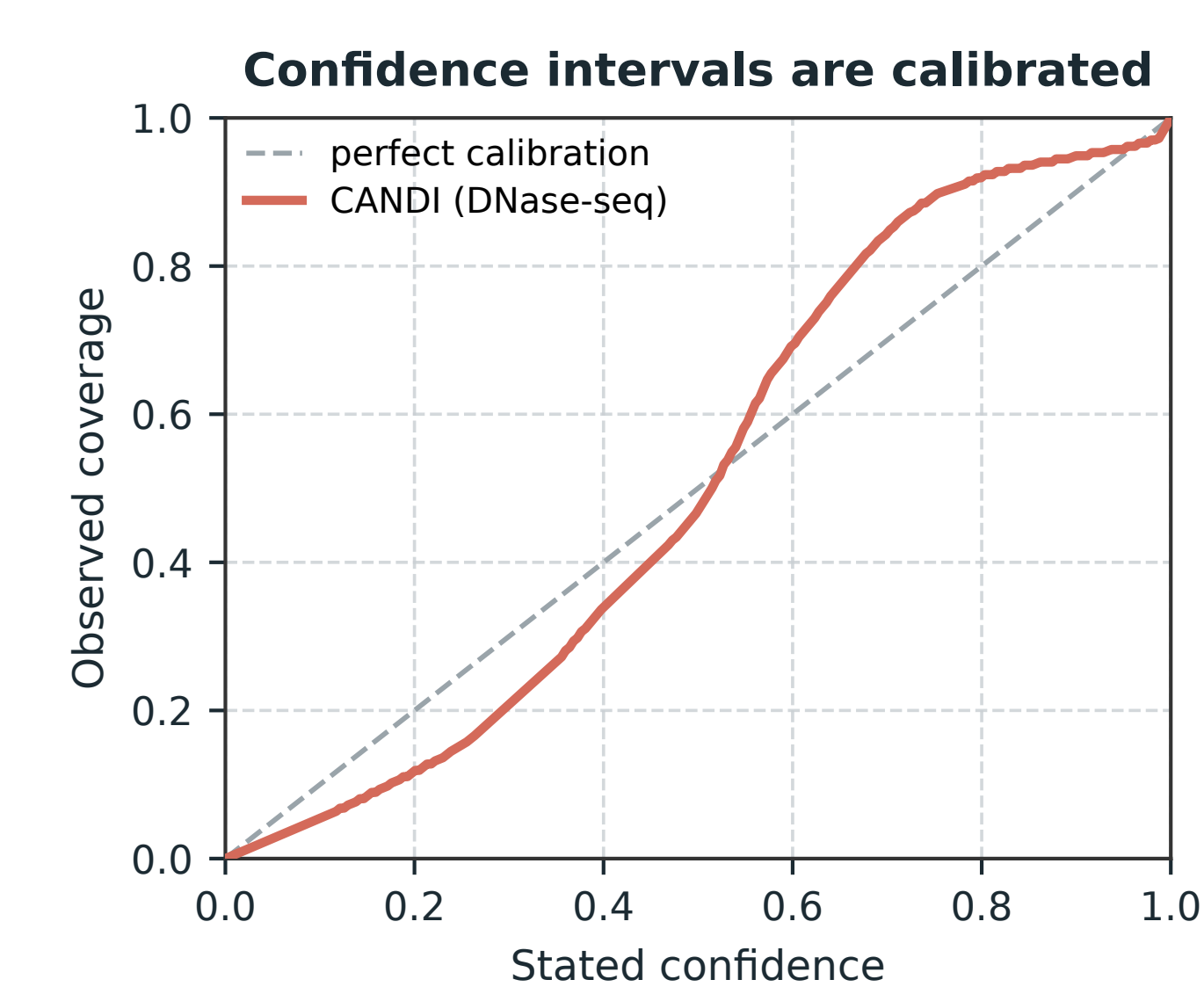


**B** Imputation score before (filled) and after (open) a correction that rescales each method's prediction and blends in the baseline. Every method improves; after it, CANDI is the best.

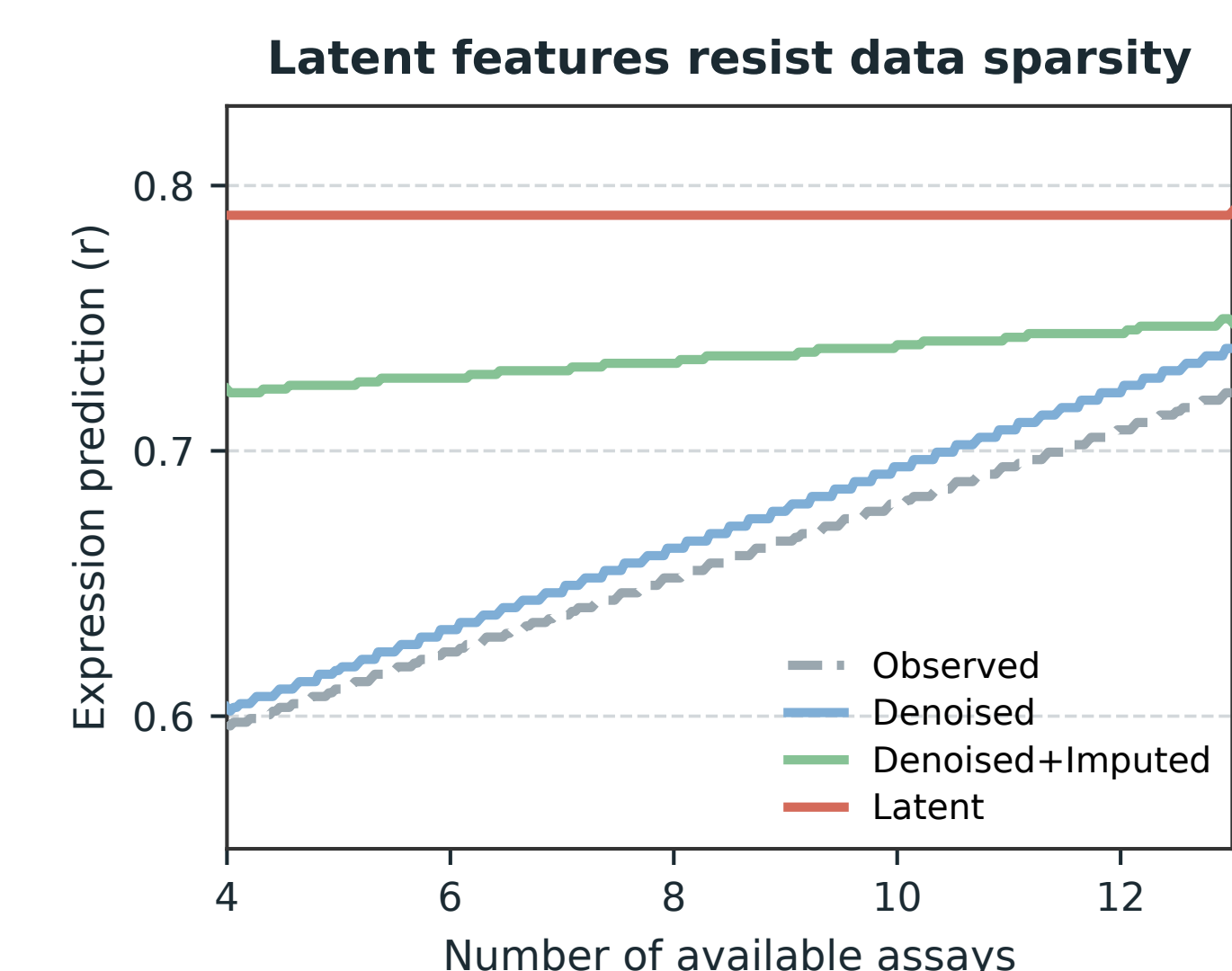


**C** Seven of the nine measures relative to the baseline, median over 51 experiments. CANDI is first on three and top 8 on the rest.

## Downstream utility of CANDI



**D** Confidence calibration for DNase-seq (DND-41): predicted confidence level against the fraction of observed data within that interval.



**E** Gene expression prediction as input assays are removed. Denoised+Imputed and Latent settings are robust to input sparsity.

[1] Ernst J, Kellis M. Large-scale imputation of epigenomic datasets for systematic annotation of diverse human tissues. *Nat Biotechnol* 33:364–376 (2015).

[2] Schreiber J, Boix C, et al. The ENCODE Imputation Challenge: a critical assessment of methods for cross-cell type imputation of epigenomic profiles. *Genome Biol* 24:79 (2023).

[3] Foroozandeh M, Diab AR, Libbrecht M. CANDI: self-supervised, confidence-aware denoising imputation of genomic data. *bioRxiv* 2025.01.23.634626 (2025).

Panels A–C: blind-test scoring from the Libbrecht lab's EIC reproduction; *baseline* is the challenge's average-activity track. CANDI's output layer is fitted on training experiments only; the correction in B is fitted on chr22 of each blind-test experiment and scored on the other 22 chromosomes.